Winter term 2020/2021: Exercises "Radio Frequency Technology"

EXERCISE SESSION 11 (29.01.2021)

Problem 11.1: Cylindrical dielectric resonator

Distinguish between the $HEM_{11\delta}$, $TM_{01\delta}$ and $TE_{01\delta}$ resonance modes of cylindrical dielectric resonator in terms of

a) transverse field distribution …… to be discussed
b) excitation techniques …… to be discussed
c) applications ……. to be discussed

Problem 11.2: Basics of ferrites

A LHCP RF magnetic field of $\vec{H} = (0.5\hat{x} + j0.5\hat{y})$ A/m is applied to a calcium vanadium garnet (CVG) ferrite medium having a saturation magnetization of $4\pi M_s = 1780$ G. Ignoring loss, calculate the resulting magnetic flux density $\vec{B}$ at f = 10 GHz for two cases:

(a) no magnetic bias field and ferrite demagnetized ($M_s = H_0 = 0$), and
(b) a z-directed magnetic bias field of 1000 Oe.

Problem 11.3: Wave propagation in ferrites

An infinite lossless ferrite medium with a saturation magnetization of 4πMs = 1200 G and a dielectric constant of 10 is biased to a field strength of 500 Oe. If a linearly polarized wave at 8 GHz is propagating in this material, what is the distance it must travel in order that its polarization is rotated 90°?

Short Questions session 11

1. A certain lossless circulator has a return loss of 10 dB. What is the isolation? What is the isolation if the return loss is 20 dB?